MATHEMATICS

Analysis and approaches

Higher Level | Set B | Paper 3

Mock examination · 2025 series

Time allowed: 1 hour 15 minutes Maximum mark: 55

Candidate name: _____

Candidate number: _____

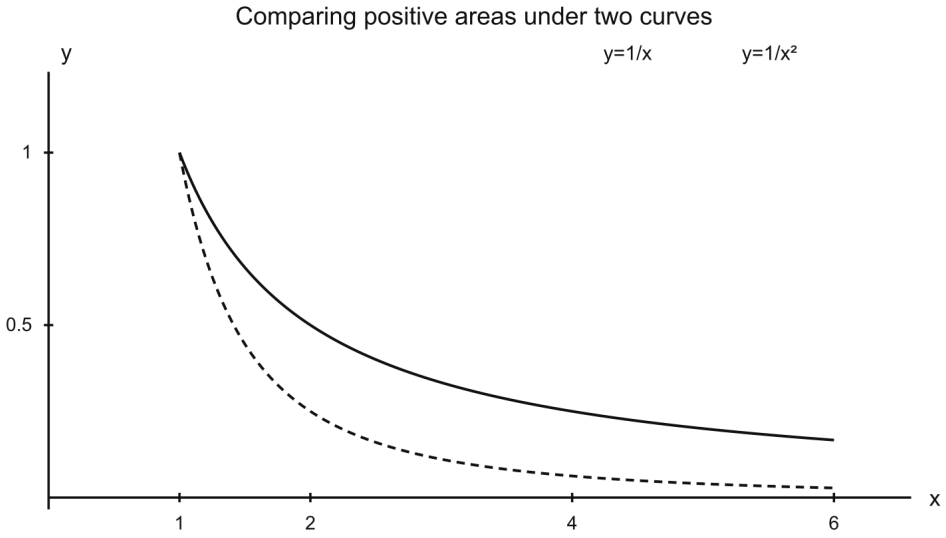
Date: _____

Instructions

- A graphic display calculator is required.
- This paper contains two compulsory extended problem-solving questions.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

Question 1 [Maximum mark: 30]

This question investigates convergence of improper integrals. Unless another restriction is given, $n > 1$ and all logarithms are natural. An improper integral is interpreted as a limit of finite integrals, not by substituting infinity into an antiderivative.



(a)(i)) Use your calculator to evaluate $\int_1^{100} \frac{dx}{x}$ and $\int_1^{1000} \frac{dx}{x}$ to three significant figures. [1]

(a)(ii)) Evaluate $\int_1^{100} x^{-2}\,dx$ and $\int_1^{1000} x^{-2}\,dx$. [1]

(b)(i)) Show that $\int_1^{n} x^{-3}\,dx = \frac{n^2-1}{2n^2}$. [2]

(b)(ii)) Hence evaluate $\int_1^{\infty} x^{-3}\,dx$, explaining the limiting step. [2]

(c)(i)) Find $\int_1^{n} x^{-1/2}\,dx$ in terms of n. [3]

(c)(ii)) Determine whether $\int_1^{\infty} x^{-1/2}\,dx$ converges. [1]

(d)(i)) For a real parameter $p \neq 0$, find $\int_1^{n} x^{-(p+1)}\,dx$. [3]

(d)(ii)) For $p > 0$, find $\int_1^{\infty} x^{-(p+1)}\,dx$ and justify convergence. [2]

(d)(iii)) State what happens to this improper integral when $p < 0$. [1]

(d)(iv)) Evaluate $\int_1^{\infty} x^{-2025}\,dx$. [1]

(e)(i)) For $x > 1$, differentiate $y = \ln(\ln x)$. [1]

(e)(ii)) For $n > 2$, evaluate $\int_2^{n} \frac{dx}{x \ln x}$. [2]

(e)(iii)) Determine whether $\int_2^{\infty} \frac{dx}{x \ln x}$ converges. [1]

(f)(i)) For $x > 1$ and $p \neq 0$, use $u = \ln x$ to find $\int \frac{dx}{x(\ln x)^{p+1}}$. [3]

(f)(ii)) For $n > 2$, evaluate $\int_2^{n} \frac{dx}{x(\ln x)^{p+1}}$. [1]

(f)(iii)) For $p > 0$, evaluate the corresponding improper integral from 2 to infinity. [1]

(f)(iv)) State what happens when $p < 0$. [1]

(g)(i)) For $p > 0$, find $\int e^{-px}\,dx$. [1]

(g)(ii)) Evaluate $\int_0^n e^{-px}\,dx$. [1]

(g)(iii)) Hence evaluate $\int_0^\infty e^{-px}\,dx$. [1]

Question 2 [Maximum mark: 25]

A fair six-sided die has its faces labelled $1, 2, 3, 5, 6, m$, where the mean of the six labels is 3.5. In the first game, players P_1 and P_2 take turns, starting with P_1. A player wins immediately on rolling the face labelled m; otherwise the next player rolls. All throws are independent. Unless a different game is specified, play continues until someone wins.

(a) Find m. [1]

(b)(i)) In the first game, find the probability r that both players fail in one complete round. [1]

(b)(ii)) Find each player\'s probability of eventually winning. [3]

(c)(i)) A third player P_3 joins and turns proceed $P_1, P_2, P_3, P_1, \ldots$, with the same winning rule. Find the probability that P_1 wins on their second turn. [1]

(c)(ii)) Find the probability that P_1 eventually wins in the three-player game. [4]

(d) Generalize the probability that P_1 eventually wins to n players taking turns with the same die rule. [1]

(e) A different two-player game gives P_1 a success probability $1/3$ on each of their turns and P_2 a success probability $1/2$ on each of theirs. Turns still alternate, with P_1 first. Find the probabilities that P_1 wins on their first, second and third turns, and show that the game is fair. [6]

(f) In a three-player game the success probabilities on each turn are $1/x$, $1/(x-1)$ and $1/(x-2)$, respectively, with $x > 3$. Find P_1\'s probability of eventually winning. [5]

(g) Now let n players have success probabilities $p_i = 1/(x - i + 1)$ for $i = 1, \ldots, n$. Assuming every p_i lies strictly between 0 and 1, prove by contradiction that $p_1 < 1/n$. [3]

End of question paper

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

1. [Maximum mark: 30]

(a) (i) $\int_1^{100} \frac{1}{x}\,dx = 4.61$, $\int_1^{1000} \frac{1}{x}\,dx = 6.91$ **A1**

(ii) $\int_1^{100} \frac{1}{x^2}\,dx = 0.99$, $\int_1^{1000} \frac{1}{x^2}\,dx = 0.999$. **A1**

[2 marks]

(b) (i) $\int_1^n \frac{1}{x^3}\,dx = \left[\frac{x^{-2}}{-2}\right]_1^n = \left[-\frac{1}{2x^2}\right]_1^n = -\frac{1}{2n^2} + \frac{1}{2} = \frac{n^2-1}{2n^2}$. **A1 A1**

(ii) $\lim_{n\to+\infty} \frac{n^2-1}{2n^2} = \frac{1}{2}$, so $\int_1^{\infty} \frac{1}{x^3}\,dx = \frac{1}{2}$. **A1 A1**

[4 marks]

(c) (i) $\int_1^n \frac{1}{\sqrt{x}}\,dx = \left[2\sqrt{x}\right]_1^n = 2\sqrt{n} - 2$ **M1 A1 A1**

(ii) $\int_1^{\infty} \frac{1}{\sqrt{x}}\,dx = +\infty$. **A1**

[4 marks]

(d) (i) $\int_1^n \frac{1}{x^{p+1}}\,dx = \int_1^n x^{-p-1}\,dx = \left[\frac{x^{-p}}{-p}\right]_1^n = \left[-\frac{1}{px^p}\right]_1^n = -\frac{1}{pn^p} + \frac{1}{p}$ **M1 A1 A1**

(ii) For $p > 0$, $\int_1^{\infty} \frac{1}{x^{p+1}}\,dx = \frac{1}{p}$ **M1 A1**

(iii) For $p < 0$, $\int_1^{\infty} \frac{1}{x^{p+1}}\,dx = +\infty$ **A1**

(iv) $\int_1^{\infty} \frac{1}{x^{2025}}\,dx = \frac{1}{2024}$. **A1**

[7 marks]

(e) (i) $\frac{dy}{dx} = \frac{1}{x \ln x}$. **A1**

(ii) $\int_2^n \frac{1}{x(\ln x)} dx = \left[\ln(\ln x)\right]_2^n = \ln(\ln n) - \ln(\ln 2)$ **M1 A1**

(iii) $\int_2^{+\infty} \frac{1}{x(\ln x)} dx = +\infty$. **A1**

[4 marks]

(f) (i) $\int \frac{1}{x(\ln x)^{p+1}} dx$

Let $u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x} \Rightarrow dx = x du$ **M1**

$\int \frac{1}{x(\ln x)^{p+1}} dx = \int \frac{1}{xu^{p+1}} x du = \int \frac{1}{u^{p+1}} du = -\frac{1}{pu^p} + c = -\frac{1}{p(\ln x)^p} + c$ **A1 A1**

(ii) $\int_2^n \frac{1}{x(\ln x)^{p+1}} dx = -\frac{1}{p(\ln n)^p} + \frac{1}{p(\ln 2)^p}$ **A1**

(iii) For $p > 0$, $\int_2^{+\infty} \frac{1}{x(\ln x)^{p+1}} dx = \frac{1}{p(\ln 2)^p}$ **A1**

(iv) For $p < 0$, $\int_2^{+\infty} \frac{1}{x(\ln x)^{p+1}} dx = +\infty$ **A1**

[6 marks]

(g) $\int \frac{1}{e^{px}} dx = \int e^{-px} dx = \frac{e^{-px}}{-p} + c = -\frac{1}{pe^{px}} + c$ **A1**

$\int_0^n \frac{1}{e^{px}} dx = \left[-\frac{1}{pe^{px}}\right]_0^n = -\frac{1}{pe^{pn}} + \frac{1}{p}$ **A1**

$\int_0^{\infty} \frac{1}{e^{px}} dx = \frac{1}{p}$ **A1**

[3 marks]

2. [Maximum mark: 25]

(a) $m = 4$. **A1**

[1 mark]

(b) (i) $r = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$ **A1**

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(ii) For P_1, $\frac{\frac{1}{6}}{1-\frac{25}{36}} = \frac{\frac{1}{6}}{\frac{11}{36}} = \frac{6}{11}$ **M1 A1**

For P_2, $\frac{5}{11}$ **A1**

[4 marks]

(c) (i) $\left(\frac{5}{6}\right)^3 \frac{1}{6}$ **A1**

(ii) 1st attempt: $\frac{1}{6}$, 2nd attempt : $\left(\frac{5}{6}\right)^3 \frac{1}{6}$, …

Geometric sequence with $u_1 = \frac{1}{6}$, $r = \left(\frac{5}{6}\right)^3$ **M1 A1**

$\frac{\frac{1}{6}}{1-\left(\frac{5}{6}\right)^3} = \frac{36}{91}$ **M1 A1**

[5 marks]

(d) $\frac{\frac{1}{6}}{1-\left(\frac{5}{6}\right)^n}$. **A1**

[1 mark]

(e) 1st attempt: $\frac{1}{3}$ **M1**

2nd attempt: $\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)=\frac{1}{9}$ **A1**

3rd attempt: $\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)=\frac{1}{27}$

G.S. with $u_1=\frac{1}{3}$, $r=\frac{1}{3}$ **M1 A1**

Probaility P_1 winning the game **overall**: $\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{\frac{1}{3}}{\frac{2}{3}}=\frac{1}{2}$, so the game is fair. **A1 A1**

[6 marks]

(f) G.S. with $u_1=\frac{1}{x}$, **M1 A1**

$r=\left(1-\frac{1}{x}\right)\left(1-\frac{1}{x-1}\right)\left(1-\frac{1}{x-2}\right)=\left(\frac{x-1}{x}\right)\left(\frac{x-2}{x-1}\right)\left(\frac{x-3}{x-2}\right)=\frac{x-3}{x}=1-\frac{3}{x}$ **M1 A1**

$S_\infty=\frac{\frac{1}{x}}{1-1+\frac{3}{x}}=\frac{\frac{1}{x}}{\frac{3}{x}}=\frac{1}{3}$ **A1**

[5 marks]

(g) $p_1=\frac{1}{x}$, $p_2=\frac{1}{x-1}$, $p_3=\frac{1}{x-2}$, $\ldots$, $p_n=\frac{1}{x-(n-1)}$

Assume, for contradiction that $p_1=\frac{1}{x}\geq\frac{1}{n}$ **M1**

Then $x\leq n\Rightarrow x-n\leq 0\Rightarrow x-(n-1)\leq 1$ **A1**

$\Rightarrow p_n\geq 1$, contradiction, **A1**

[3 marks]